

# Exploratory Data Analysis (EDA)

## Python Code Solution

Below is the properly structured and corrected Python code for your Exploratory Data Analysis project. Several OCR errors from the original text (such as missing = signs, missing commas in figsize, incorrect formatting in the include parameter, incorrect f-string brackets, and lack of indentation in the loop) have been repaired to ensure the code executes perfectly.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv("data.csv")

# Display basic dataset information
print(df.head())
print(df.shape)
print(df.columns)
print(df.dtypes)
print(df.describe())

# Check for missing and duplicate values
print(df.isnull().sum())
print(df.duplicated().sum())

# Correlation Matrix Visualization
corr = df.select_dtypes(include=np.number).corr()

plt.figure(figsize=(8, 6))
plt.imshow(corr, cmap="coolwarm")
plt.colorbar()
plt.xticks(range(len(corr.columns)), corr.columns, rotation=90)
plt.yticks(range(len(corr.columns)), corr.columns)
plt.title("Correlation Matrix")
plt.tight_layout()
plt.show()

# Histogram Visualization
df.hist(figsize=(12, 8))
plt.suptitle("Histogram of Numerical Features")
plt.show()

# Boxplot Visualization for numeric columns
numeric_cols = df.select_dtypes(include=np.number).columns

for col in numeric_cols:
    plt.figure(figsize=(6, 4))
    plt.boxplot(df[col].dropna())
    plt.title(f"Boxplot of {col}")
    plt.show()
```

```
# Final status messages
print("EDA Completed Successfully!")
print("- Checked missing values")
print("- Analyzed statistical summary")
print("- Found correlations")
print("- Visualized data distributions")
```